Shellfs

Read and write shell pipelines through file paths

Install & load

```
SET allow_extension_repositories = 'allowed';
CREATE EXTENSION REPOSITORY IF NOT EXISTS query_farm WITH PREFIX 'https://duckdb-extensions.query.farm';
INSTALL shellfs FROM query_farm;  LOAD shellfs FROM query_farm;
```

01 At a glance

Use a shell command as a DuckDB file source or sink. A trailing pipe reads command output; a leading pipe sends COPY output to a command. Useful for local tools that already produce or consume CSV and JSON.

Local command → Streaming pipe → DuckDB rows

02 Pipe paths + SQL integration

`'command |'`

Read the command's standard output as a file.

`'| command'`

Write bytes to the command's standard input.

`read_csv(pipe_path, ...)`

Parse streamed CSV with normal DuckDB reader options.

`read_json(pipe_path, ...)`

Parse JSON emitted by a command.

`COPY (...) TO pipe_path`

Stream query results into a command; specify the format.

`SET ignore_sigpipe = true`

Ignore SIGPIPE; a LIMIT that stops reading early is never an error.

`SET shellfs_enabled = false`

One-way switch that turns shell commands off; SET GLOBAL for all connections.

03 Keep in mind

Pipe placement sets direction

Place | at the end to read, or at the beginning to write; the wrong direction is an error. Commands run through the host shell with the process's permissions, and none run while enable_external_access is false. Never build commands from untrusted input.

04 Query a command's CSV output

```
SELECT sum(n) AS total
FROM read_csv(
  'printf "n\n1\n2\n3\n" |',
  header = true
);
-- 6
```

Runs local printf; the trailing pipe makes it a readable path.

05 Compress an export with gzip

```
COPY (SELECT range AS n FROM range(1, 4))
TO '| gzip > /tmp/qf-shellfs.csv.gz'
(FORMAT CSV, HEADER, COMPRESSION UNCOMPRESSED);
```

Requires gzip and writable /tmp. Disable COPY compression so only gzip compresses the stream.

06 Read the archive back

```
SELECT sum(n) AS total
FROM read_csv('/tmp/qf-shellfs.csv.gz');
-- 6
```

Run after the export above; DuckDB detects gzip from the filename.

07 In practice

Treat the source as a stream

Pipes do not behave like seekable local files. CSV and newline-delimited JSON are natural fits. Queries waiting on a pipe can be cancelled, and a nonzero exit after DuckDB reads all output is an error. Ensure the command and its tools exist on the host.

Full docs & examples ↗ Source ↗